

Melinda A. Tran

+1 (971) 570-6892 | meltran@bu.edu | [linkedin.com/in/melintran](https://www.linkedin.com/in/melintran) | github.com/MelindaTan | melindatan.github.io/PortfolioWebsite/

EDUCATION

Boston University — College of Engineering

B.S. Computer Engineering, Computer Science Minor, Machine Learning Concentration

Expected May 2027

Relevant Coursework: Digital logic, Applied Algorithms, Signals and Systems, Computer Organization, Computer Networking, Machine Learning, etc.

Member: SWE, uWISE, Theta Tau, High Performance Computing Club, EngBraguids

TECHNICAL SKILLS

Programming: UI/UX, C++, Git, Python, Java, JavaScript, HTML, CSS, MATLAB, HDL

Robotics: Pytorch, Tensorflow, OpenCV, Vuforia, ROS1, ROS2, CVAT, Roboflow, MoCap

Platforms: React, React Native, Angular, Figma, Linux

CAD: NX, PTC Creo, PolyWorks, SolidWorks, Onshape

WORK EXPERIENCE

Red Hat

Incoming Software Engineer Intern

Boston, MA

Jun 2025 - Aug 2025

Break Through Tech + Cornell University

ML & AI Fellow

Ithaca, NY

Mar 2026 - Present

- Selected (from 4000 applicants) for the competitive, year-long initiative focused on machine learning, data science, and applied AI
- Built and evaluated machine learning models using real-world datasets, applying data preprocessing, training, and validation techniques
- Collaborated on industry sponsored AI projects, working in teams and receiving mentorship from industry professionals to develop technical skills

BU Robotics Lab

Computer Vision Undergraduate Researcher

Boston, MA

Oct 2025 - Present

- Integrate neuromorphic maps (sight) with autonomous control systems to automate complex robot manipulation tasks
- Engineered high-fidelity robotic control systems using Motion Capture (MoCap) data, managed via the Robot Operating System (ROS)

BIT Lab

LLM Undergraduate Researcher

Boston, MA

Jun 2025 - Nov 2025

- Researched LLM memory, reasoning, focusing on architecture, training objectives, advanced prompting, fine-tuning, and memorization dynamics
- Analyzed the impact of training and optimization strategies on LLM problem-solving, memorization, and generalization capabilities

AgAID + Cornell University

Machine Vision Research Intern

Ithaca, NY

May 2025 - Aug 2025

- Train and test AI models using CVAT (**100+ Images**) to detect obstacles and develop navigation paths for robotic systems in dynamic environments
- Integrated models with robotic systems using ROS2, OpenCV, and Roboflow for real-world testing in apple orchards with **86% accuracy**
- Apply AI-driven data processing for traversability path detection, integrate models with robotic systems to assess performance in commercial agricultural settings

Metakosmos

Computer Engineer Intern

Sydney, NSW

Feb 2025 - May 2025

- Conducted in-depth research and testing of AI-driven manufacturing optimization strategies, implementing pilot programs that enhanced operational efficiency, reduced production costs, and improved scalability through workflow automation and intelligent modeling tools
- Assessed real-time metrics like energy use, uptime, and resource efficiency using DSG, TRL, and MRL frameworks to inform engineering strategy

Weir ESCO

Photogrammetry Design Engineer Intern

Portland, OR

Jun 2022 - Aug 2022

- Received hands-on training in NX and PolyWorks metrology software for 3D modeling and dimensional analysis
- Applied material-science principles to design and run stress tests on components, before and after production, evaluating durability and performance
- Partnered with the MEOT team to analyze photogrammetry scans of mining-equipment parts (points, adapters, shrouds), then 3-D modeled and printed prototypes to present and pitch design improvements

PROJECTS

Burger Bash Animated Game | C++ | Java |

Nov 2024 - Dec 2024

- Led a team to develop a fully functional 2D animated game by integrating front-end and back-end components for a smooth user experience
- Designed and implemented key game mechanics using C++ and Java, deploying the application on Android Studio for mobile compatibility
- Debugged, optimized, and refactored code with unit tests using Agile to improve performance, stability, and long-term maintainability

First Tech Challenge Robotics | Java |

Jan 2021 - Jan 2022

- Developed machine learning (ML) capabilities using object detection tools such as OpenCV, Vuforia, and TensorFlow to enhance robotic vision
- Programmed both TeleOp and autonomous robot functions, integrating rotation and light sensors to enable responsive and self-directed navigation

LEADERSHIP

Theta Tau

Director of Website Development

Boston, MA

May 2025 - Present

- Built and deployed a fully responsive website using HTML, CSS, and JavaScript tailored to the needs of the BU Engineering Fraternity
- Integrated a GitHub repository with Cloudflare Pages for seamless version control, automatic deployment, and real-time updates
- Configured custom domain, DNS settings, and SSL certificates through Cloudflare to ensure secure, stable, and high-performance web hosting